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Injection Well Stimulation: Putting Away More For Less In California Waterfloods

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Abstract

The waterflood is the most widely used and most successful secondary recovery technique in the oil industry. The injection of produced or make-up water into an oil reservoir can increase recoveries from an average of 19% of original oil in place to as much as 55 % [1] by driving or sweeping additional oil from the reservoir as the water cycles from injection wells to production wells.

The challenge to operators of water flood fields is to maximize their water injection rates at as low a cost as possible since injection wells themselves are not revenue-generating assets. Just as production wells periodically require stimulation to improve productivity, the injection well also develops near-wellbore skin damage that reduces injectivity over time. Injection water quality and incompatibilities with native reservoir fluids can lead to scale or other debris blocking pore throats and reducing near-wellbore permeability – thus requiring stimulation.

A large percentage of California injection wells are completed across multiple sands. Although fracture stimulation treatments generally result in greater conductivity between the wellbore and the reservoir, acid stimulation remains the technique of choice for restoring the injectivity of the majority of these wells. A fracture treatment would result in communication of these sands behind casing and eliminate the ability of the operator to control which sands are taking injection water.

This paper discusses unique acidizing chemistry and placement techniques that have demonstrably improved injectivity while maintaining the integrity of the barriers between sands in a number of waterflood fields in California. Optimization of acid chemistry and placement improves the depth of penetration of the treatment, reduces the risk of near-wellbore formation deconsolidation, extends treatment life, and reduces the overall cost per barrel of water injected.

Introduction

Waterflood systems in the oil producing sandstone reservoirs of the Los Angeles Basin and San Joaquin Valley are a crucial part of field development. They provide pressure maintenance, prevent subsidence, sweep additional oil from the injection wells to production wells and provide convenient disposal of produced water.

To improve development economics, injection wells in a water flood are often completed across several oil bearing zones rather than drilling a separate injector for each interval. They can be perforated at the zones of interest, or completed with long slotted liners to maximize the length of formation face taking water. Just like production wells, injection wells require maintenance to maximize injection volumes and rates at minimal cost.

Skin damage from scale deposits and dirty injection water can reduce permeability in the near wellbore area resulting in high injection pressures and reduced injection rates. High pump pressures on an injection system result in higher energy and maintenance costs and shortens equipment life. Pressure limitations of the injection system eventually result in lower injection rates as the pressure drop through the damage zone rises. Lower rates create produced water disposal problems, reduced sweep efficiency and reduced oil production.

The solution to this is some form of stimulation. Acidizing has become the preferred method of restoring a wells injectivity, typically measured as

$$\textbf{Injectivity} = \frac{\textbf{Injection rate}}{\textbf{Injection pressure}}$$

in barrels of water per day per psi of injection pressure. Damage to injection wells tends to be a near wellbore phenomenon with the critical skin radius being less than 2 feet as illustrated in **Figure 1**. The reduction in injectivity is a direct function of the loss in average permeability using Darcy's radial flow equations.

$$\frac{\textbf{Damaged Injectivity}}{\textbf{Undamaged Injectivity}} = \frac{\textbf{Damaged Average Permeability}}{\textbf{Undamaged Permeability}}$$

$$\frac{I_1}{I_0} = \frac{\ln(r_e / r_w)}{\ln(r_e / r_1) + K_e * K_1^{-1} * \ln(r_1 / r_w)}$$

Low rate acid stimulation treatments are near wellbore matrix damage removal techniques aimed at restoring radial permeability. A fracture treatment could be used to bypass this type of damage but is undesirable due to reasons previously mentioned, in addition to being quite a bit more costly. As a waterflood field ages and the flood front reaches the production wells, water breakthrough can lead to thief zones where water is unprofitably cycled through the reservoir.

Effective diversion techniques must be employed to prevent a stimulation treatment from being wasted on a thief zone. For multizone injection wells, the diversion and placement techniques that are effective in matrix acid treatments cannot be applied to control the placement of a fracture treatment. A considerable risk in fracturing an injection well is connecting all of the injection points to one thief zone, drastically reducing the effectiveness of that particular injection well.

This paper will discuss common damage mechanisms in water injection wells, commonly employed acid systems, acid placement techniques, the economics of waterflood stimulation, and show two case histories where innovations in placement techniques and acid chemistry have helped to inject more water at a lower cost per barrel.

Damage Mechanisms

Water quality of the injected fluid plays a critical role in sustaining the injection rate of water injection wells for both case histories. Generally, lower permeability reservoirs are especially prone to injectivity reduction due to water quality problems. Specific metrics that define water quality include total suspended solids (TSS), particle size distribution of the suspended solids, and the amount and type of dissolved minerals in the injection water (hardness).

The location where these metrics are measured is important, since water quality is dynamic as it travels through the injection system from the water processing plant, where it is usually treated and filtered, to the injection well perforations. Old and decaying distribution lines can easily change clean water at the treatment plant to a damaging solids laden fluid by the time it reaches the reservoir. Perforation plugging and permeability reduction can occur when particle size exceeds one-third to one-half of the reservoir pore throat diameter. If filtration of the injection water does not reduce entrained particle diameter to less than this threshold, skin damage begins to develop.

The compatibility of injection water and formation or produced water is an important part of minimizing damage to injection wells. Formations that are sensitive to fresh or low salinity water can be severely damaged due to the swelling of clays in the rock matrix which can restrict or totally shut-off the effective pore space of the injection zone.

Scale deposits can result when dissolved minerals in the injection water react with minerals in the formation and the formation brine. Typical scales of this type include calcium carbonate ($CaCO_3$), calcium sulphate ($CaSO_4$), barium sulphate ($BaSO_4$) and iron carbonate (Fe_2CO_3). Barium is the most difficult to remove. Geochemical analyses of both waters can determine the likelihood and severity of such reactions (scale index). An injection water with a high concentration of dissolved minerals can precipitate out solids when a change in temperature or pressure occurs. One example is iron oxide (Fe_2O_3), which can be dissolved from the flow lines but can precipitate out at the formation face.

A dominant plugging agent in waterfloods is iron sulphide slime, a byproduct of the metabolic process of sulphate-reducing bacterium (SRB). Iron sulphide and the bio-mass of the SRB builds up on the internal surfaces of iron pipe and can be transported with the injection water into the perforations where plugging (and decreased injectivity) can occur. Effective biocide treatment is essential to control the growth and presence of SRB in injection water. Internal coatings of injection tubing and line pipe is also an effective deterrent to SRB growth because contact of the injection water with iron is minimized, thus eliminating the food source of the SRB. Once the damage is in place, however, acidizing is the only way to remove it.

The suspected source of the damage mechanism in the injection well must be identified prior to acidizing. This allows the correct treatment fluid to be selected. If possible, the source of the damage should be corrected prior to the stimulation treatment to prevent the damage from reoccurring. In some cases though, it is less costly to acidize every 6 months or yearly than it is to upgrade a water treatment plant or replace a distribution system.

Acid Systems

The two most common acid systems employed in the oilfield today are still hydrochloric acid (HCl) and hydrofluoric acid (HF). Each system can be applied in a variety of strengths depending on the type of rock and the targeted damage mechanism. In each case, a variety of chemical additives are blended with the acid for corrosion protection, prevention of iron precipitation or emulsions and reduction of friction or surface tension. This paper evaluates only the effect of differing acid systems on similar reservoirs as the additive packages remain consistent.

In a sandstone reservoir, HCl is generally only effective when targeting calcium and iron based scales or plugging material such as bacteria and polymer gel in a well.[3] It does not dissolve siliceous fines or clays which may be plugging pores. HCl is used in strengths ranging from 5% to 28% although 10% or 15% concentrations are most common.

The selection of acid concentration is based on several factors. At higher temperatures the concentration may be reduced without significantly affecting performance as the rate of reaction of the acid increases with increase in temperature. A high concentration of HCl acid will spend so rapidly at high temperature that effective depth of penetration may not be any greater than a lower concentration which reacts more slowly.[3] In a case such as this, the lower cost of a weaker acid system can govern the choice of fluid. A large volume of soluble scale in the wellbore or perforations may require the greater dissolving power of a stronger acid, while the risk of formation deconsolidation may favor a weaker concentration if the cementatious material of the sandstone is soluble in HCl.

Temperature and economics were the primary considerations governing the choice of acid strength in the following case histories. Another benefit of HCl systems is that they generally do not generate precipitates as a byproduct of reactions in the wellbore or formation in an injection well. Caution should be exercised if there will be oil present in the near wellbore because live HCl can react with some crudes to create damaging sludges and solids. In a typical waterflood, the acid is spent and extremely dilute long before it reaches the oil bearing rock out in the reservoir.

HF acid systems are used to dissolve silicates and clays located in the near wellbore rock matrix and to remove migrating formation fines. Neither are soluble in HCl acid alone. In well stimulation, HF acid is typically created by combining ammonium fluoride (AF) or ammonium bifluoride (ABF) with hydrochloric acid. Mixtures of the two acids are prepared by diluting mixtures of the concentrated acids with water or by adding fluoride salts to hydrochloric acid. The fluoride salts release HF acid when dissolved in HCl acid. Concentrations of HF in HCl solutions range from 0.5 percent to 9 percent.[2]

In some of the following case histories, 9% HCl : 1% HF and 12% HCl : 3% HF mud acid mixtures were used. Unfortunately, HF acid does not react solely with silicates. In order of preference, HF reacts with carbonates, clays and then silicates. This preference for non-siliceous material can result in all of the HF acid being spent before it is able to remove it's intended target, the silicates. To offset this, an HCl, citric, formic or acetic acid preflush must be used ahead of HF acid to remove the majority of carbonate material and allow the HF to react with the silicates and clays.

Formations rich in clay content often require some type of delayed reaction (retarded) acid system to achieve the necessary depth of penetration of live acid. Another disadvantage of mud acid systems is that HF reactions can generate insoluble secondary byproducts which precipitate as the post treatment pH rises. This is addressed by a HCl or similar post flush to depress the pH until the spent acid can be flowed back out of the well or over displaced far from the critical near wellbore region. For injection well stimulation the simplest solution is to put the well on injection immediately following the final acid or overflush stage to push the stimulation fluids far out into the reservoir.

An alternative chemistry to the conventional HCl-HF acid systems was also employed in both case histories in an effort to improve the net increase in injectivity and extend the life of the treatments. In the improved chemistry system, HCl is replaced with a phosphonic acid complex (HV acid). The HV acid is reacted with the AF or ABF to produce HF. Because the hydrogen ions contained on the HV acid molecule dissociate at different stochiometric conditions, the HF which has already been generated will react and only then will more HF be produced. This self generating HF acid system reduces the rate at which the acid system can react, allowing increased depth of penetration of live HF acid into the formation. The initial reaction of the HV:HF system with clays creates a thin aluminum silicate phosphonate coating on the clay, preventing further spending of the HF. By reducing the reaction rate on high surface area clays, the volume of actual silica that can be dissolved is increased. This improves near wellbore permeability.

Furthermore, as the HF is produced, an ammonium phosphonate salt is evolved from the HV Acid which remains in solution. This is less likely to create insoluble precipitates as the pH of the spent acid system rises compared to conventional HF acid systems. There is also improved wettability with this HV:HF chemistry, resulting in improved acid contact with the material targeted to dissolve.

The result of this slower overall reaction rate is deeper penetration into the formation and reduced risk of formation deconsolidation in the near wellbore area, as seen in the compressive strength tests of **Figure 3**. The compressive strength at the beginning, middle, and end of a section of core is a measure of the reactivity of the acid at that point. Note that a conventional HF treatment nearly deconsolidates the formation at the wellbore while several inches away the strength of the rock matrix is untouched.

Strong chelating and dispersing properties of the HV:HF acid system prevent the unwanted precipitation of fluorosilicates, hexafluorosilicates, alumino-fluorosilicates, iron compounds and calcium fluoride, all of which can be liberated from scale and formation rock during acidizing with conventional HF.[4,5] This sequestrating property reduces the need for a low pH system found in conventional HF blends where a 4:1 HCl-HF ratio is used to prevent precipitation. The lower overall acid strength (same dissolving power, higher pH) of the HV:HF system reduces the requirement for corrosion inhibitors in high temperature environments. A 12% HCl – 3% HF conventional system requires 0.8% of a common corrosion inhibitor for 6 hours of protection while the HV:HF system requires just 0.4%. This can lead to considerable cost savings. These properties of the improved HV:HF acid system are in part responsible for the improved results seen in numerous case histories.[6]

Placement Techniques

A number of different placement techniques have been tried to determine which approach yields the best increase in injectivity as well as which technique generates the most economic results. To summarize, bullhead treatments were performed with and without diversion and coiled tubing treatments were performed with and without diversion. A list of the techniques and the corresponding number of treatments performed are found in **Table 1.**

The completions in the majority of the injection wells are long perforated intervals or long sections of slotted liner covering a number of oil-bearing zones. Scale and debris from the injection water plug off the perforation tunnels as well as gradually block the slots in the liner. Periodic spinner surveys illustrate the decline in injectivity as well as the growth of thief zones in some intervals (**Figure 2**).

A variety of diversion techniques were tested over several years in an effort to maximize the percentage of each wellbore that is treated. Solid diverters such as benzoic flakes and calcium carbonate were used. Foam and annular water injection were used to divert acid away from thief zones. The use of coiled tubing creates a second flow path that can be used to place acid below a thief zone while injection water is pumped into the thief down the annular flow path. If the thief is low in the interval, the same effect can be achieved by pumping the acid down the annulus and the diversion water down the coiled tubing. Spinner surveys are necessary to optimize this simple technique.

Coiled tubing and a variety of jetting tools were used for mechanical diversion. The number of jobs covered by these case histories (174) is large enough that some statistically significant trends emerge. The data in **Table 2** shows that on average, a treatment placed using coiled tubing will show 60% greater improvement than a bullheaded treatment.

The coiled tubing success can be attributed to controlling the placement of the treatment fluids across the entire interval. A bullheaded treatment will initially take the path of least resistance until acid contacts the first interval, then most of the treatment that follows will go to that same zone. Heavily damaged intervals may never be exposed to the treatment fluid if insitu benign wellbore fluids cannot be pushed out of the way. One may even speculate that bullhead treatments have a greater chance of creating thief zones due to non-uniform delivery of the acid.

In an attempt to further enhance the delivery of acid to the damaged intervals, a controlled speed rotating jetting tool (R-Jet) was placed on the end of the coiled tubing for a number of jobs. Coiled tubing and a static nozzle were used as a baseline for comparison on previous treatments on the same wells with the same treatment fluid.. As shown in **Table 3,** even with the same type of acid on the same well, use of the R-Jet tool resulted in a greater gain in injectivity and a smaller decline over a 3 month period.

The advantage of the R-Jet nozzle lies in the ability to remove blockages from all perforation tunnels and all slots in a liner where as a static nozzle will only direct acid toward targets that pass directly in front of it. By opening up the entire interval to water injection, greater initial rates are possible and it requires a longer period of time for subsequent damage to accumulate to the point that injectivity is adversely affected.

Economics

Water injection wells are not direct revenue generating assets in the way that an oil producing well is. For this reason, the economics of injection well stimulation must be evaluated a bit differently. The cost of water disposal must be considered. The impact of sweep efficiency and higher reservoir pressure on the field's oil production must also be determined.

Another factor which is difficult to quantify is the cost of subsidence damage to wellbores. In some fields, if the volume of oil produced is not replaced by water (voidage) the compressibility of the formation as the pore pressure declines can result in ground shifts that can collapse or shear off casing strings. Water disposal costs in fields without water injection wells can range from $0.50 per barrel to as much as $4.00 per barrel depending on transportation costs and the location of the field relative to the disposal site. As long as the net value of an incremental barrel of oil (sales price less production cost and stimulation cost) is positive, there is a good case for injection well stimulation.

The cost of incremental oil production attributed to the waterflood stimulations in one of the case histories is illustrated in **Figure 4**. Repair of subsidence collapsed casing and proper abandonment of unrepairable wells can easily exceed $100,000.00 and a replacement well must still be drilled to recover lost production at a cost of $75,000 to $400,000 depending on well configuration. From this perspective, injection well stimulation can be considered preventative maintenance and an affordable method of lowering field operating costs.

An economic injection well stimulation involves more than just minimizing the cost of the treatment. As the following case histories show, increasing the average cost of a treatment by 40% through the use of coiled tubing for placement increased the average injectivity increase by 60%. **Figure 5** and **Figure 6** illustrate that switching to a slightly more expensive acid chemistry gave a greater increase in injectivity and a slower decline rate. This reduces the frequency of stimulations required to maintain injection volumes. If the cost of incremental oil produced is above a certain cutoff point (varies from flood to flood, **Figure 4**), a more expensive treatment may be more economical. A $30,000 treatment performed once every 6 months has a lower annual cost than a $20,000 treatment performed once every 3 months. Care must be taken not to focus solely on cost per treatment.

Case History: Coastal California Water Flood Field

The coastal California injection wells are part of a waterflood in a tightly folded, severely faulted anticline structure in the Los Angeles Basin. The injecton pattern varies by fault block. The field consists of multiple interbedded sandstones and mudstone sequences with varying permeabilities. The producing sands are Pliocene age turbidite sands ranging in depth from 2600 ft down to 14,000 ft with 1000 to 3000 feet of pay exposed iin many wells.[8] First put into primary production in 1926, the waterflood was first implemented in the late 1950's. Many of the injection wells are completed across multiple sands by perforations or slotted liners. Placement techniques are a significant part of optimizing treatments on these wells to prevent stimulation fluids from entering thief zones. **Table 1** lists the various methods used. The improved acid chemistry (HV:HF acid) was the sole variable evaluated on 3 wells. The use of a controlled speed rotating jetting nozzle (R-Jet) was the sole variable changed on three wells.

The target intervals in the three wells used to evaluate the improved HV:HF acid system are at 10,000 ft with temperatures between 180 F and 230 F. These wells were previously acidized several times with a design consisting of 3000 to 4000 gal of 10% HCl, a similar volume of 9% HCl : 1% HF and the same volume of ammonium chloride overflush. Clay content in these formations is fairly low so higher HF concentrations are unnecessary.

Design volume was kept approximately the same when evaluating the improved HV:HF acid system. The acid design was again using 10% HCl as a preflush, the equivalent of a 1% HF system, and an ammonium chloride overflush. In all three cases, the improved HV:HF acid system yielded greater initial increases in injectivity and slower decline rates, as illustrated in **Figure 5** and **Figure 6**. The injectivity increase averaged 50% higher and the decline rate dropped in half.

Previous work determined that placement of stimulation fluids using coiled tubing is more effective than simple bullheading as shown in **Table 2**. The coiled tubing technique averaged a 62% increase in injectivity improvement with longer treatment life. To further optimize placement efficiency, the use of a controlled speed rotating jetting nozzle

(R-Jet) was the sole variable changed on an additional three wells. These wells were stimulated with a 15% HCl acid system several times. The most recent treatment involved the R-Jet nozzle. Improvements were seen in both injectivity increase and longevity of treatment. This is thought to be due to focused placement in the perforations and slots and full 360 degree coverage to maximize the surface area of the formation taking injection fluid. The effects of the nozzle are shown in **Table 3**.

Three other wells in **Table 3** were treated with both the improved HV:HF acid system and the rotating nozzle. With two variables changed it is not possible to determine the results due to each individual factor. The results, however, clearly indicate that injectivity can be increased and decline rate slowed by the use of optimized chemistry and placement. Not all of the 17 sands under water flood showed positive economics. **Figure 4** shows the cost of stimulation per incremental barrel of oil recovered for the 5 sands that responded the best to injector stimulation. This data is used to rank the importance of future stimulation candidates.

Case History: San Joaquin Valley Waterflood Field

The San Joaquin Valley injection wells are part of a pattern waterflood in a Late Miocene Stevens anticline structure. The depositional environment is a turbidite fan system. The sands are stratified which leads to a non-uniform water flood front. Originally set up as a peripheral flood in 1978, the system was changed to a pattern flood in 1996 to recover bypassed oil due to the non-uniform flood front.[7] Wells are on a 10 to 20 acre spacing. The four wells shown in **Figure 7** are at an average depth of 7,500 ft and an average temperature of 215 F with a formation thickness of about 400 ft. Porosity is in the range of 18 to 25%.

Treatment design volumes for the conventional HF acid systems averaged 21 gal/ft 15% HCl preflush, 14 gal/ft 12% HCl:3% HF and 28 gal/ft 9% HCl:1% HF for the main stage and an overflush with 14 gal/ft of ammonium chloride. The HCl preflush was designed to remove any carbonates, the strong HF acid blend was designed to remove damage from the near wellbore, and the weaker HF acid blend was designed to penetrate further into the formation. No HCl overflush was used; instead, all treatment fluids were overflushed back into the reservoir where any precipitates would not impact the radial permeability near the wellbore.

The design for the improved HV:HF acid system was slightly different. The preflush was 23 gal/ft of 10% HCl, the main stage was 23 gal/ft Regular Strength HV:HF acid blend (a 3% HF equivalent in terms of HF concentration) followed by 23 gal/ft 10% HCl and an 18 gal/ft ammonium chloride overflush before putting the well immediately on injection. With a reservoir temperature over 200 F it was felt that 10% HCl would be sufficient to remove any carbonates (scale etc.) from the near wellbore.

The improved HV:HF acid system yielded significant improvement in initial injectivity response and a reduced decline rate as illustrated in **Figure 7**. The conventional HF system averaged a three fold improvement in injectivity with a 50% decline over seven months. The improved HF acid system averaged a seven-fold improvement in initial injectivity with a decline of 15% over the same time period. A larger number of candidates should be compared to prove up these statistics, but when taken with similar results in other waterfloods in California, Nigeria, Indonesia and South America, the benefits of the improved HV:HF acid system cannot be denied.

<u>Conclusion</u>

Waterflood operating costs can be reduced and associated oil production increased by optimizing water flood stimulations. The following conclusions can be drawn from the data and case studies:

1. Some type of diversion and coiled tubing placement are required for uniform treatment and to prevent stimulation fluids from entering thief zones.
2. The HV:HF acid chemistry improved the stimulation of the sandstone formations providing greater initial increases in injectivity and slower decline rates than previous acid systems.
3. The controlled speed rotating jetting tool (R-Jet) provides better results than a static jet nozzle when all other factors remain the same.
4. Decisions on the economic viability of a stimulation treatment should be made based on cost per incremental barrel of oil produced, not just on initial treatment cost.

Nomenclature

BHA	= *Bottom hole assembly or downhole tools*
BPM	= *Barrels per minute*
CTU	= *Coiled Tubing Unit*
gpm	= *U.S. gallons per minute*
HCl	= *hydrochloric acid*
HF	= *hydrofluoric acid*
HV	= *phosphonic acid*
K_{avg}	= *average radial permeability*
K_e	= *undamaged permeability*
K_1	= *damaged permeability*
r_e	= *drainage radius*
r_1	= *damage radius*

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SI Conversion Factors

bbl	x 161.7	E + 00 L
°F	x 1.8 +32	°C
US gal	x 3.785	L
psi	x 6.8948	kPa

Placement Technique	Average Cost (CDN)	Injectivity (bpd/psi)			Life
		Pre Job	Post Job	Net	(months)
Bullhead	$19,300	0.28	0.9	0.62	9.75
CTU	$27,000	0.24	1.25	1.01	10.5

TABLE 2. Results of bullheading vs coiled tubing, 174 jobs

Technique	Number of Jobs
Bullhead, no diversion	18
Bullhead, solid diversion	23
Bullhead, foam diversion	10
CTU jet back w N2	4
CTU cleanout	9
CTU cleanout & stim, no diversion	18
CTU, no diversion	66
CTU, solid diversion	2
CTU, water diversion	4
CTU, foam diversion	6

TABLE 1. Number of injection well treatments

Placement Technique	Fluid System	Injectivity (bpd/psi)	
		Gain	Decline
Static Nozzle	HCl	1.035	0.292
R-Jet nozzle	HCl	1.115	0.186
Static Nozzle	HF	0.101	0.0968
R-Jet Nozzle	HF	0.185	0.0282

TABLE 3. Effect of coiled tubing nozzle selection

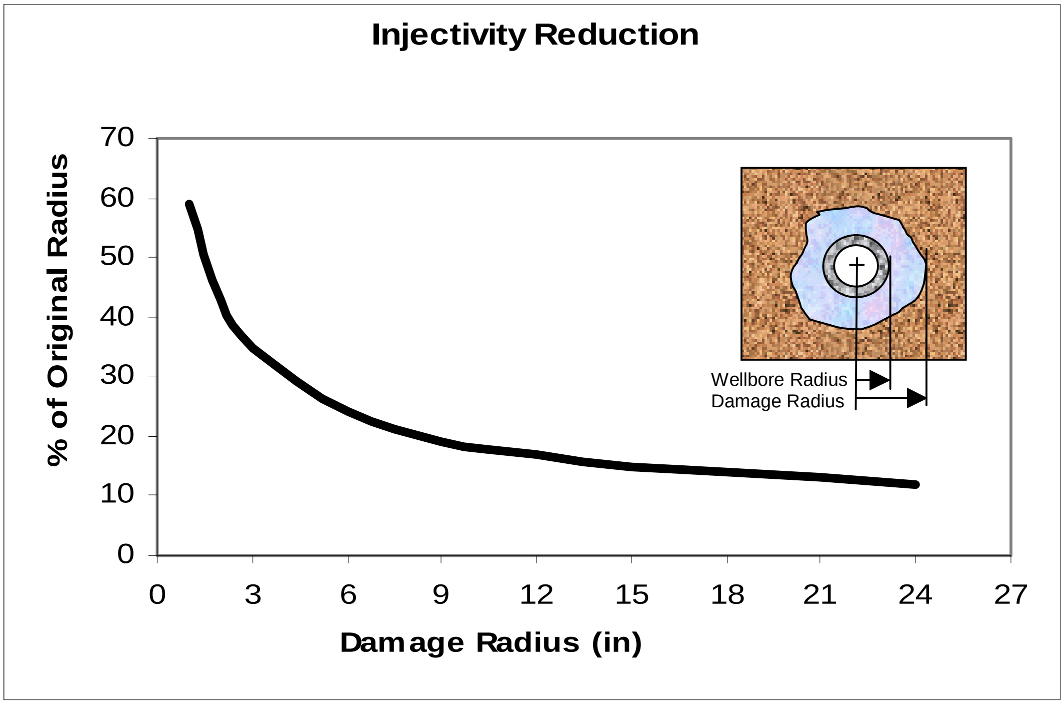

Figure 1. Relationship between injectivity and damage radius

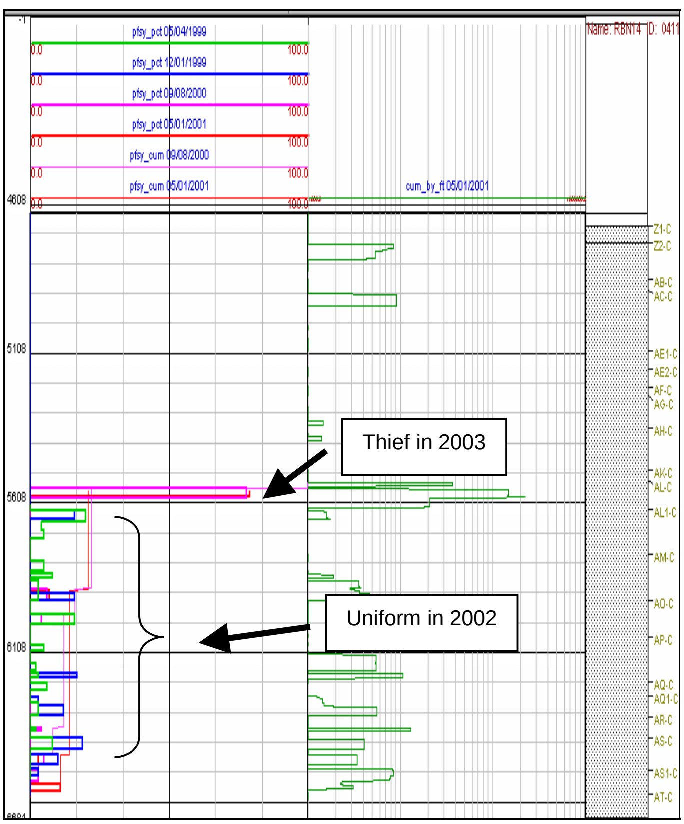

Figure 2. Development of thief zone.

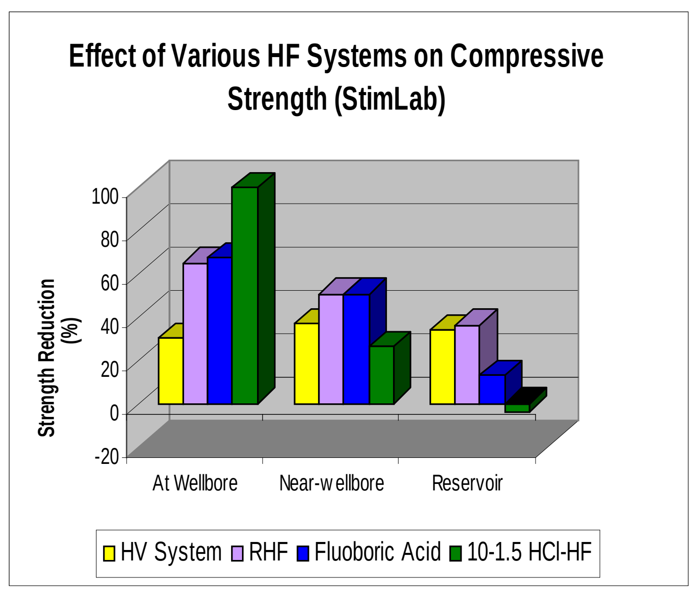

Figure 3. Live penetration depth of various HF acid systems

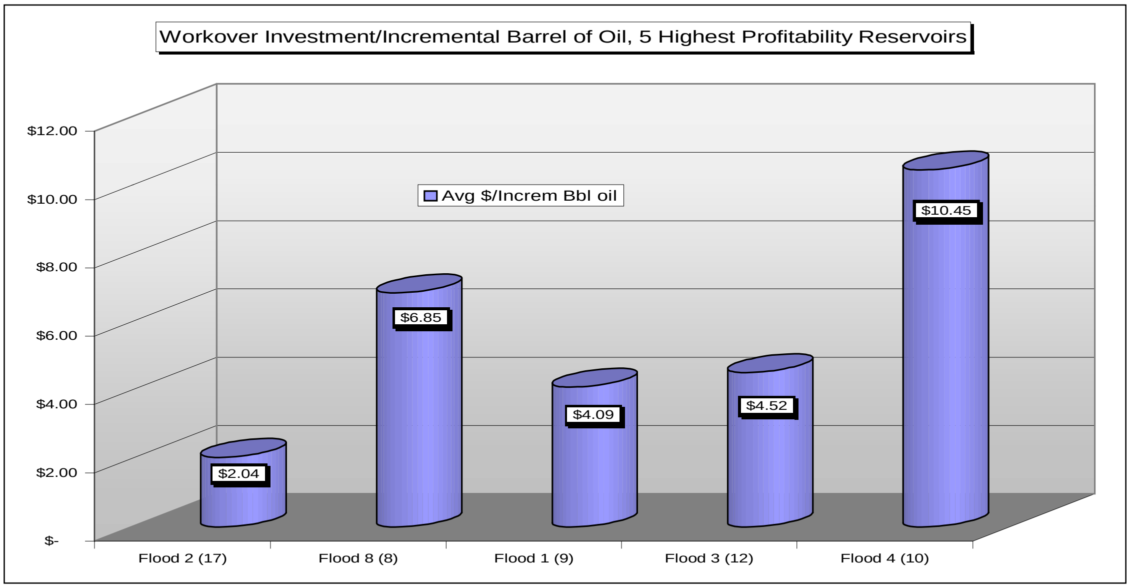

Figure 4. Value of injection well workovers

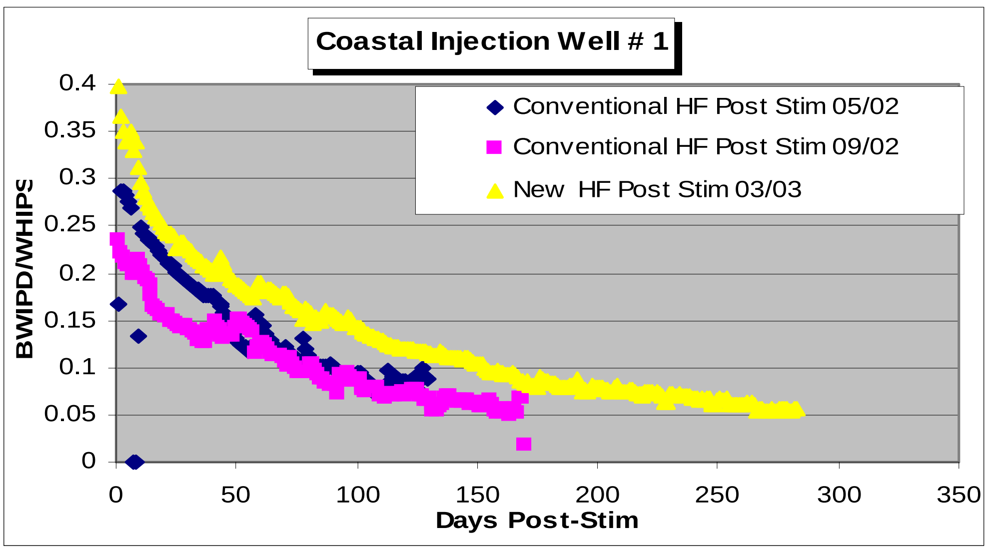

Figure 5. Effect of various acid systems on Well #1 injectivity

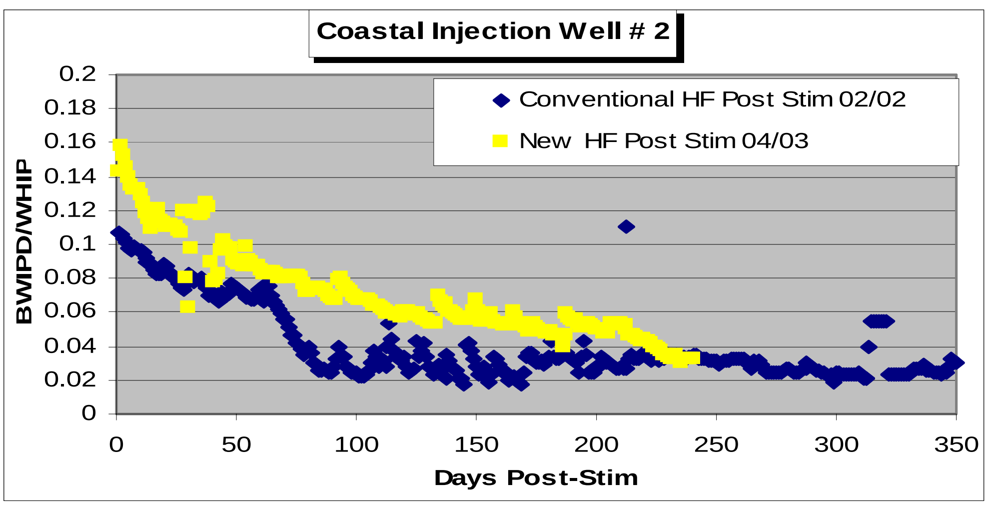

Figure 6. Effect of various acid systems on Well #2 injectivity.

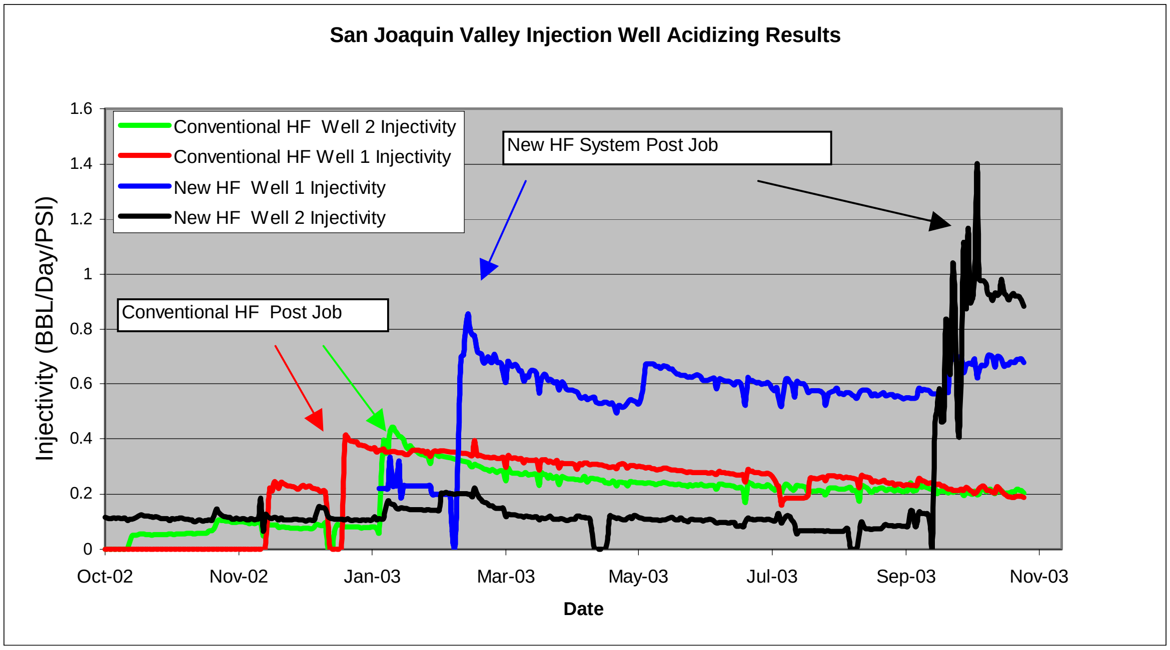

Figure 7. Comparison of conventional and new HF systems